Quillfeather

EXECUTIVE SUMMARY

Quillfeather Inc. is a registered C Corporation founded by three students at our university. We make an innovative handwriting practice notebook with a sensor pen and a companion app that helps elementary school children improve their penmanship and keeps parents informed about progress. This document gives an overview of our problem, solution, market, business model, competition, team, milestones and what we are asking the competition for.

THE PROBLEM

Handwriting is a foundational skill for children in the early grades. Studies that we have read suggests handwriting is linked to reading, spelling and idea generation, though we admit the research is mixed on how strong the link is. Despite this, teachers in many schools have less and less classroom time for penmanship, because of testing, tablets and a crowded curriculum. Parents often want to help at home but they do not know what good handwriting looks like at each grade level, they cannot tell if the child is forming letters the right way (for example starting a letter from the bottom instead of the top), and worksheets give no feedback when the parent isn't sitting next to the child. In our early interviews with about twenty parents and four teachers, most said they worried about handwriting but had no good tool, and several said they simply gave up on practice at home because it turned into arguments.

OUR SOLUTION

Quillfeather is a system of three parts working together. First is a notebook with pages printed with guided letter and word exercises designed around common grade level standards. Second is a sensor pen that looks and feels like a normal chunky pencil-style pen but contains motion and pressure sensors and a small chip that records how each letter is written. Third is a companion app that receives the data by Bluetooth and turns it into feedback. While the child writes, the app shows encouraging prompts and small games, and flags things like wrong stroke direction, uneven size, or pressing too hard. After each session it saves a record, and every week the parent gets a progress report showing practice time, areas improving, and what to work on next. We believe this takes the guessing out of home practice and makes it more fun for kids, which they asked for in our testing.

MARKET

There are roughly 4 to 5 million children in each elementary grade in the United States, so around 25 million children in grades K through 5. Our initial target is children in grades 1 through 3, about 12 million, whose families spend money on educational supplies, workbooks and learning apps. If just 1 percent of these children's families bought one kit that would be over 100,000 customers, which at our planned price is a meaningful business. We recognise that this is a top down estimate and we have not yet done a detailed bottom up analysis of how many families will actually pay for a device for handwriting. Secondary markets we may look at later include schools, tutors, occupational therapists and homeschool groups, as well as international markets.

BUSINESS MODEL

We make money in three ways. The starter kit, which includes the sensor pen, one notebook and app access, sells for 49 dollars. Refill notebooks sell for 12 dollars for a pack of two. Parents can also subscribe to the premium app features, such as detailed reports and multiple children profiles, for 5 dollars per month. Our estimated cost to make the kit is currently around 21 dollars at small volumes, which we think will come down with larger orders, and refills cost about 4 dollars to print. We plan to sell mainly through our website and online marketplaces first, then later through retailers and school supply channels. Our customer acquisition plan relies on social media, parent bloggers, and teacher word of mouth, although we haven't yet tested paid advertising so we do not know our real acquisition costs.

COMPETITION

There are many ways parents help children with handwriting. Traditional workbooks and worksheets are cheap but give no feedback. Several handwriting apps for tablets exist where children trace letters with a finger or stylus, but these keep children on screens and do not work with real paper, and many do not give parent reporting. There are also general smart pens aimed at adults for note taking and digitizing, which are more expensive and not designed for children's learning. Our advantage is that children write on real paper, with immediate feedback, in a product designed for kids and reporting designed for parents. We know that larger companies could copy our idea if it works, and our main protection right now is speed and the design of our feedback engine, for which we have not yet filed a patent but are considering it.

TEAM

Marisol Tran-Okafor is our CEO, a senior studying business with a minor in education, who ran the parent and teacher interviews and leads sales and partnerships. Dev Aldenhoven is our CTO, a computer engineering student who designed the pen electronics and the first version of the firmware. Priya Sandoval-Reyes is our head of product and software, studying computer science, who built the app and the stroke analysis. We do not yet have anyone with deep experience in manufacturing at scale or in childrens education product compliance, and we are looking for advisors in these areas.

MILESTONES AND FINANCIALS

So far we have defined the core concept, built a working prototype of the pen, notebook and app, and incorporated the company as a C Corp. We have placed the prototype with 14 families in a first pilot, and children used it for about six weeks on average. In the pilot, parents reported that they liked the weekly reports, though a few said the pen was a bit heavy and that Bluetooth pairing was annoying. One customer has paid us 49 dollars for a unit, which is our first revenue, and three more families have said they would buy but have not yet. We have spent approximately 6,800 dollars so far, funded by the founders and a small university grant. We have a website with a waitlist. Over the next twelve months we project selling about 1,500 kits, and revenue of roughly 100,000 dollars including refills and subscriptions, though these numbers are early estimates and depend heavily on whether we can fix the pen and get cheaper manufacturing.

WHAT WE ARE ASKING FOR

We are applying to this competition to win prize funding, which we would use to produce a better second version of the pen that is lighter and pairs more easily, to make a first production run of around 500 units, to run a bigger pilot with at least 60 families and two classrooms, and to pay for legal help on safety testing for childrens products and a possible patent filing. Beyond money, we would value mentorship from judges with experience in hardware, childrens products and consumer marketing, and introductions to manufacturers and to schools. We believe Quillfeather is a real opportunity to help children with an important skill and we thank the judges for considering our application.